

Markov Operator

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Chapter 1

tmp

1.1 Markov Process and Semigroups

1. **Markov Process:** Let E be a Polish space that is a separable complete metric space and let E be equipped with the Borel σ -field $\mathcal{F}$. Then the *measure decomposition theorem* implies that for any probability measure μ on the product σ -field $\mathcal{F} \otimes \mathcal{F}$ on $E \times E$ with $\mu_1 = \pi_1^\# \mu$, the first projection, then

$$\mu(dx, dy) = k(x, dy)\mu_1(dx)$$

for some probability transition kernel $k: E \times \mathcal{F} \rightarrow [0, 1]$. Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space.

Remark 1.1.1. Moreover, because of the existence of kernels, by Ionescu–Tulcea theorem, for any probability measure μ on E^n , there are kernels k_i from E^{i-1} to E such that

$$\mu(dx_1, dx_2, \dots, dx_n) = \mu_1(dx_1)k_2(x_1, dx_2)k_3(x_1, x_2, dx_3) \cdots k_n(x_1, \dots, x_{n-1}, dx_n).$$

Let $(X_t^x)_{t \geq 0}$ be a E -valued Markov process with $X_0^x = x \in E$, or denoted by $(X_t)_{t \geq 0}$. Let $\mathcal{F}_t = \sigma(X_u; u \leq t)$ be the natural filtration of $(X_t)_{t \geq 0}$. So the *Markov Property* implies that

$$\mathbb{E}[X_t | \mathcal{F}_s] = \mathbb{E}[X_t | X_s], \quad \forall t > s$$

Moreover, here we consider the *Time-Homogeneous Markov Process*, that is

$$\mathbb{E}[X_t | X_s] = \mathbb{E}[X_{t-s} | X_0]$$

For any X_t , consider the pair (X_t, X_0) , let $p_t(x, dy)$ be the corresponding transition probability kernel. Note that if $X_0 = x$, then $p_t(x, dy)$ is also the distribution. Because of the time-homogeneity, for any pair (X_{t_1}, X_{t_2}) with $t_1 \leq t_2$, the transition probability measure

$$p_{t_1 \rightarrow t_2}(y_1, dy_2) = p_{0 \rightarrow t_2 - t_1}(y_1, dy_2) = p_{t_2 - t_1}(y_1, dy_2)$$

Therefore, for $X_0 = x$, the distribution of (X_{t_1}, X_{t_2}) is

$$p_{t_1}(x, dy_1)p_{t_2 - t_1}(y_1, dy_2)$$

More generally, for $0 < t_1 \leq \dots \leq t_k$, the distribution of $(X_{t_1}, \dots, X_{t_k})$ given that $X_0 = x$,

$$p_{t_1}(x, dy_1)p_{t_2 - t_1}(y_1, dy_2) \cdots p_{t_k - t_{k-1}}(y_{k-1}, dy_k)$$

Remark 1.1.2. For $(X_{t_1}, \dots, X_{t_k})$, its distribution $p_{1\dots k}^x$ of $X_0 = x$ has the decomposition

$$p_{1\dots k}^x(dy_1, \dots, dy_k) = p_1^x(dy_1)k_2(y_1, dy_2)k_3(y_1, y_2, dy_3) \cdots k_k(y_1, \dots, y_{k-1}, dy_k)$$

by Ionescu–Tulcea theorem. First, $p_1^x(dy_1) = p_{t_1}(x, dy_1)$. $\mathbb{E}[X_t \mid \mathcal{F}_s] = \mathbb{E}[X_t \mid X_s]$ implies that

$$k_i(y_1, \dots, y_{i-1}, dy_i) = p_{t_{i-1} \rightarrow t_i}(y_{i-1}, dy_i) = p_{t_i - t_{i-1}}(y_{i-1}, dy_i).$$

Moreover, the Markov process is uniquely determined by the finite-dimensional distributions of X_t , so the transition probability kernels $\{p_t(x, dy)\}_{t \geq 0}$ uniquely determine the Markov process. Or equivalently, we define the associated Markov semigroup $(P_t)_{t \geq 0}$, that is for a suitable measurable function $f: E \rightarrow \mathbb{R}$,

$$P_t f(x) := \mathbb{E}[f(X_t^x)] = \mathbb{E}[f(X_t) \mid X_0 = x] = \int_E f(y) p_t(x, dy)$$

and $P_0 f = f$. It is a true semigroup, because for any $t, s \geq 0$,

$$\begin{aligned} P_{t+s} f(x) &= \mathbb{E}[f(X_{t+s}^x)] = \mathbb{E}[\mathbb{E}[f(X_{t+s}^x) \mid \mathcal{F}_t]] \\ &= \mathbb{E}[\mathbb{E}[f(X_{t+s}^x) \mid X_t^x]] \\ &= \mathbb{E}[P_s f(X_t^x)] = P_t(P_s f)(x) \end{aligned}$$

where the first equality in the third line is because the time-homogeneity

$$\begin{aligned} \mathbb{E}[f(X_{t+s}) \mid X_t = x_t] &= \int_E f(y) p_{t \rightarrow t+s}(x_t, dy) \\ &= \int_E f(y) p_s(x_t, dy) \\ &= (P_s f)(x_t) \end{aligned}$$

Therefore, $P_{t+s} = P_t \circ P_s$. $(P_t)_{t \geq 0}$ acting on measurable functions induces the dual semigroup $(P_t^*)_{t \geq 0}$ acting on measures on E , that is

$$\int_E f d(P_t^* \nu) := \int_E P_t f d\nu$$

Or more directly, for any $A \in \mathcal{F}$,

$$P_t^* \nu(A) = \int_E p_t(x, A) d\nu(x),$$

i.e., $(P_t^* \nu)(dx) = \int_E p_t(y, dx) \nu(dy)$. On the other words, if $X_0 \sim \nu$, then $X_t \sim P_t^* \nu$.

2. **Semigroups:** Consider a family $\mathbf{P} = (P_t)_{t \geq 0}$ of operators acting on some set of real-values measurable functions on $(E, \mathcal{F})$, $\mathbf{P}$ satisfies the following properties:

- (i) $P_t: B(E, \mathcal{F}) \rightarrow B(E, \mathcal{F})$ is linear and transforms bounded measurable functions to bounded measurable functions;
- (ii) (*Initial condition*) $P_0 = Id$;
- (iii) (*Mass conservation*) $P_t \mathbf{1} = \mathbf{1}$ for constant function $\mathbf{1}(x) \equiv 1$;
- (iv) (*Positivity preserving*) for $f \geq 0$, $P_t f \geq 0$
- (v) (*Semigroup property*) for every $t, s \geq 0$, $P_t \circ P_s = P_{t+s}$

Corollary 1.1.3 (Jensen's inequality). *For any convex $\phi: \mathbb{R} \rightarrow \mathbb{R}$ and any $f \in B(E, \mathcal{F})$,*

$$P_t(\phi(f)) \geq \phi(P_t f)$$

Proof. Because ϕ is convex, for any $b \in \mathbb{R}$, there is $a = a(b)$ such that

$$\phi(c) \geq \phi(b) + a(b)(c - b), \quad \forall c \in \mathbb{R}$$

For any $x \in E$, let $c = f(x)$. We have

$$\phi(f(x)) \geq \phi(b) + a(b)(f(x) - b) \Rightarrow \phi(f) \geq \phi(b) + a(b)(f - b)$$

By the positivity and mass properties of P_t , we have

$$P_t(\phi(f)) \geq \phi(b) + a(b)(P_t f - b)$$

So for any $x \in E$,

$$P_t(\phi(f))(x) \geq \phi(b) + a(b)(P_t f(x) - b)$$

Then let $b = P_t f(x)$, we get

$$P_t(\phi(f))(x) \geq \phi(P_t f(x))$$

which is true for any $x \in E$. □

In particular, for $\phi(r) = |r|^p$ ($1 \leq p < \infty$), we have $P_t(|f|^p) \geq |P_t f|^p$.

Definition 1.1.4. Given a family $\mathbf{P} = (P_t)_{t \geq 0}$ on a measurable space $(E, \mathcal{F})$, a (positive) σ -finite measure μ in $(E, \mathcal{F})$ is said to be invariant for $\mathbf{P}$ if for any bounded positive function $f: E \rightarrow \mathbb{R}$ and every $t \geq 0$,

$$\int_E P_t f d\mu = \int_E f d\mu.$$

In other words, $P_t^* \mu = \mu$ for any $t \geq 0$, i.e., $\int_E p_t(y, dx) \mu(dy) = \mu(dx)$.

Remark 1.1.5. By the Jensen's Inequality, if μ is invariant for $\mathbf{P}$ if for any $t \geq 0$, P_t is a contraction on the bounded functions in $L^p(\mu)$ for any $1 \leq p < \infty$. Therefore, P_t can be extended to be defined on $L^p(\mu)$ because the space of bounded functions is dense in L^p . In particular, the definition of invariance can be extended to for $f \in L^1(\mu)$.

Therefore, we further require $\mathbf{P} = (P_t)_{t \geq 0}$ satisfying:

(vi) Let μ be a σ -finite invariant measure for $\mathbf{P} = (P_t)_{t \geq 0}$. For every $f \in L^2(\mu)$, $P_t f \rightarrow f$ in $L^2(\mu)$ as $t \rightarrow 0$.

Definition 1.1.6 (Markov Semigroup). A family of operators $\mathbf{P} = (P_t)_{t \geq 0}$ defined on a state space $(E, \mathcal{F})$ with a σ -finite invariant measure μ satisfying (i)-(vi) is called a Markov semigroup of operators.

3. *Kernel for Markov semigroup:*

Proposition 1.1.7 (Kernel Representation). *Given a good measure space $(E, \mathcal{F}, \mu)$ (for example a Polish space) with μ σ -finite, let P be a linear operator sending positive measure functions to positive measurable functions. Then, if $P(\mathbf{1}) = \mathbf{1}$ and P is bounded on $L^1(\mu)$, then there is a probability kernel $p(x, A)$ such that for every bounded or positive measurable function f on E and $(\mu$ -almost) every $x \in E$,*

$$Pf(x) = \int_E f(y) p(x, dy).$$

Definition 1.1.8 (Density Kernel). A Markov semigroup $\mathbf{P} = (P_t)_{t \geq 0}$ on $(E, \mathcal{F})$ is said to admits density kernels with respect to a reference σ -finite measure m on $\mathcal{F}$ if there exists for $t > 0$ a positive measurable function $p_t(x, y)$ defined on $E \times E$ such that for every bounded and measurable function f and m -almost every $x \in E$,

$$P_t f(x) = \int_E f(y) p_t(x, y) dm(y).$$

It can further require $p_t(x, \cdot) \in L^2(\mu)$ and such $\mathbf{P}$ admits L^2 density kernel.

Proposition 1.1.9. *Given a good measure space $(E, \mathcal{F}, m)$ (for example a Polish space) with m σ -finite, let $P: L^1(m) \rightarrow L^\infty(m)$ be linear with $\|P\|_{1, \infty} \leq M$. Then, there exists a measurable $p(x, y)$ defined on $E \times E$ with $\|p\|_\infty \leq M$ such that for any $f \in L^1(m)$ and m -almost every $x \in E$,*

$$P f(x) = \int_E f(y) p(x, y) dm(y).$$

Lemma 1.1.10. *Let $(E, \mathcal{F}, m)$ be a probability space. Let $\mathcal{G} \subset \mathcal{F}$ be a sub- σ -algebra. For $f \in L^1(m)$, let $f_G = \mathbb{E}[f \mid \mathcal{G}]$. If $\mathcal{G}$ is generated by a partition $A_1, \dots, A_n$, whether*

$$f_G(x) = \sum_{j=1}^n \alpha_j \chi_{A_j}(x),$$

where

$$\alpha_j = \frac{1}{m(A_j)} \int_{A_j} f(x) dm(x).$$

Proof. By the uniqueness of conditional expectation, it is sufficient to check:

- (i) $\mathcal{G}$ -measurability: it is clear, because each χ_{A_j} is $\mathcal{G}$ -measure.
- (ii) Consistency: For any $G \in \mathcal{G}$,

$$\int_G \sum_{j=1}^n \alpha_j \chi_{A_j}(x) dm(x) = \sum_{j=1}^n \alpha_j m(A_j \cap G) = \sum_{j=1}^n \int_{A_j \cap G} f(x) dm(x) = \int_G f(x) dm(x).$$

□

4. **Chapman-Kolmogorov Equations:** Let $\mathbf{P} = (P_t)_{t \geq 0}$ be a Markov semigroup on $L^2(\mu)$ with an invariant measure μ and it admits kernels $p_t(x, dy)$. By $P_t \circ P_s = P_{t+s}$,

$$p_t(x, dy) = \int_{z \in E} p_t(z, dy) p_s(x, dz),$$

which is called a Chapman-Kolmogorov equation. Then we can construct the process $(X_t)_{t \geq 0}$ with $X_0 \sim \nu$ with the law of $(X_0, X_{t_1}, \dots, X_{t_k})$ given by

$$\mathbb{E}[f(X_0, X_{t_1}, \dots, X_{t_k})] = \int_{E^{k+1}} f(y_0, \dots, y_{t_k}) p_{t_k - t_{k-1}}(y_{k-1}, dy_k) \cdots p_{t_1}(y_0, dy_1) \nu(dy_0)$$

With extra works such that $(X_t)_{t \geq 0}$ has continuous paths, $(X_t)_{t \geq 0}$ is a corresponding Markov process.

1.2 Markov Generator

1. **Infinitesimal Generator:** In general, let $\mathcal{B}$ be a Banach space and $(P_t)_{t \geq 0}$ be linear operators $P_t: \mathcal{B} \rightarrow \mathcal{B}$ such that:

- (i) $\|P_t x\| \leq \|x\|$.
- (ii) $P_t \circ P_s = P_{t+s}$.
- (iii) $\lim_{t \rightarrow 0} P_t x = x = P_0 x$.

These also implies that $t \mapsto P_t x$ is continuous. For any bounded measure ν on $\mathbb{R}_+$,

$$P_\nu = \int_0^\infty P_t d\nu(t).$$

It follows that

$$P_\nu P_{\nu'} = P_{\nu'} P_\nu = \int_0^\infty \int_0^\infty P_{t+s} d\nu(t) d\nu'(s).$$

For any $\lambda > 0$, the resolvent operator

$$R_\lambda := \int_0^\infty P_t e^{-\lambda t} dt.$$

So

$$R_\lambda P_t x = P_t R_\lambda x = e^{\lambda t} \int_t^\infty P_s x e^{-\lambda s} ds.$$

Furthermore,

$$R_\lambda - R_{\lambda'} = (\lambda' - \lambda) R_\lambda R_{\lambda'} = (\lambda' - \lambda) R_{\lambda'} R_\lambda,$$

i.e.,

$$R_\lambda = R_{\lambda'} (\text{Id} + (\lambda' - \lambda) R_\lambda).$$

which implies that $\text{Im } R_\lambda \subset \text{Im } R_{\lambda'}$ and so $\text{Im } R_\lambda = \text{Im } R_{\lambda'} =: \mathcal{D}$. Because $\lambda R_\lambda x = \int_0^\infty P_{s/\lambda} x e^{-s} ds$ and $P_t x \rightarrow x$ as $t \rightarrow 0$, we have

$$\lim_{\lambda \rightarrow \infty} \lambda R_\lambda x = x,$$

which means that $\mathcal{D} = \text{Im } R$ is dense in $\mathcal{B}$. Fix $\lambda > 0$, let L defined on $\mathcal{D}$ by, for any $x = R_\lambda y$,

$$Lx := \lambda x - y.$$

By the formula of $R_\lambda P_t y$, we get

$$\left. \frac{d}{dt} \right|_{t=0} P_t R_\lambda y = \lambda R_\lambda y - y$$

i.e., $\partial|_{t=0} P_t x = Lx$ for any $x \in \mathcal{D}$. For any $t \geq 0$,

$$\partial_t P_t x = L P_t x = P_t Lx$$

L is called the infinitesimal generator of $\mathbf{P}$ with domain $\mathcal{D}(L) = \mathcal{D}$.

Definition 1.2.1 (Markov Generator). Let $\mathbf{P} = (P_t)_{t \geq 0}$ be a Markov semigroup on $(E, \mathcal{F})$ with invariant measure μ . The infinitesimal generator L of $\mathbf{P}$ on $\mathcal{B} = L^2(\mu)$ is called the Markov generator of $\mathbf{P}$.

Consider a Markov semigroup $\mathbf{P} = (P_t)_{t \geq 0}$ and its infinitesimal generator L with $L^2(\mu)$ -domain $\mathcal{D}(L)$. Let $\mathcal{A} \subset \mathcal{D}(L)$ be a vector subspace such that for each $f, g \in \mathcal{A}$, $fg \in \mathcal{D}(L)$.

Definition 1.2.2 (Carré du Champ Operator). Define a bilinear map $\Gamma: \mathcal{A} \times \mathcal{A} \rightarrow L^2(\mu)$ as

$$\Gamma(f, g) = \frac{1}{2} (L(fg) - fLg - gLf),$$

which is called the carré du champ operator of the Markov operator L .

Because $P_t(f^2) \geq (P_tf)^2$, in the limit as $t \rightarrow 0$, $L(f^2) \geq 2fLf$, which implies $\Gamma(f, f) \geq 0$. Let $\Gamma(f) := \Gamma(f, f)$ for $f \in \mathcal{A}$. Then, by the bi-linearity and the positivity, the Cauchy-Schwarz inequality implies

$$\Gamma(f, g)^2 \leq \Gamma(f)\Gamma(g), \quad \forall f, g \in \mathcal{A}.$$

For convex function ϕ such that $\phi(f) \in \mathcal{D}(L)$ when $f \in \mathcal{D}(L)$, Jensen's Inequality implies

$$L\phi(f) \geq \phi'(f)Lf.$$

Let $(X_t)_{t \geq 0}$ (with $X_0 = x$) be the corresponding Markov process such that $t \mapsto f(X_t)$ is right-continuous and

$$M_t^f = f(X_t) - \int_0^t Lf(X_s)ds, \quad t \geq 0,$$

is a local martingale. If it is a true martingale, then $\mathbb{E}[M_t^f] = \mathbb{E}[M_0^f]$ implies that

$$P_tf(x) = f(x) + \int_0^t P_sLf(x)ds,$$

which, from the abstract semigroup view,

$$P_t = \text{Id} + \int_0^t P_sLds.$$

Consider the Riesz functional calculus of L ,

$$Q_tf = e^{tL}f = \sum_{k=0}^{\infty} \frac{t^k}{k!} L^k f$$

then $\partial_t Q_t = LQ_t$. Therefore,

$$P_t = Q_t = e^{tL}.$$

2. **Fokker-Planck Equations:** Let $\mathbf{P} = (P_t)_{t \geq 0}$ be a Markov semigroup that admits kernels $p_t(x, y)$ with respect to reference measure m and let L be its generator. By $\partial_t P_t f = LP_t f$,

$$\partial_t p_t(x, y) = L_x p_t(x, y), \quad p_0(x, y)dm(y) = \delta_x.$$

Furthermore, because L is defined on $L^2(m)$, we can consider the its adjoint L^* also defined on $L^2(m)$ given by

$$\int_E fL^*gdm = \int gLf dm.$$

Therefore,

$$\partial_t p_t(x, y) = L_y^* p_t(x, y),$$

which is called the Fokker-Planck equation (Kolmogorov forward equation).

3. **Symmetric Markov Semigroup:** Let $\mathbf{P} = (P_t)_{t \geq 0}$ be a Markov semigroup on $(E, \mathcal{F})$ with invariant measure μ and L be the Markov operator. Let $(X_t)_{t \geq 0}$ (with $X_0 = x$) be the corresponding Markov process.

Definition 1.2.3 (Symmetric Markov Semigroup). The Markov semigroup $\mathbf{P} = (P_t)_{t \geq 0}$ is said symmetric with respect to the invariant measure μ if for any $f, g \in L^2(\mu)$,

$$\int_E f P_t g d\mu = \int_E g P_t f d\mu$$

In such case, μ is called reversible for $\mathbf{P}$.

Remark 1.2.4. (1) In terms of kernel $p_t(x, dy)$, μ -reversibility is

$$p_t(x, dy)\mu(dx) = p_t(y, dx)\mu(dy),$$

which directly implies that for any $0 \leq t \leq T$, the law of (X_t, X_T) is the same as that of (X_{T-t}, X_0) , because for $(X_t = x, X_T = y)$, its law is

$$\int_{x_0} \mu(dx_0) p_t(x_0, dx) p_{T-t}(x, dy) = \mu(dx) p_{T-t}(x, dy),$$

by the invariance of μ , and for $(X_{T-t} = x, X_0 = y)$, its law is $\mu(dy) p_{T-t}(y, dx)$. Furthermore, for any $0 \leq t_1 \leq \dots \leq t_k \leq t$, the law of $(X_{t_1}, \dots, X_{t_k}, X_t)$ is as the same as that of $(X_{t-t_1}, \dots, X_{t-t_k}, X_0)$, which means the distribution of $(X_s)_{0 \leq s \leq t}$ is as the same as that of $(X_{t-s})_{0 \leq s \leq t}$.

(2) In terms of L ,

$$\int_E f L g d\mu = \int_E g L f d\mu$$

for $f, g \in \mathcal{D}(L)$. So L is self-adjoint on $\mathcal{D}(L) \subset L^2(\mu)$.

Lemma 1.2.5 (Rota's Lemma). Let $\mathbf{P} = (P_t)_{t \geq 0}$ be a Markov semigroup on $(E, \mathcal{F})$ symmetric with respect to invariant measure μ . Then, for any $1 < p < \infty$, there exists a $C_p > 0$ such that for any measurable function $f: E \rightarrow \mathbb{R}$,

$$\left\| \sup_{s \geq 0} P_s f \right\|_p \leq C_p \|f\|_p.$$

Proof. Let f be a bounded measurable function and $M_t := P_{T-t} f(X_t)$ for $t \in [0, T]$. Then for $0 \leq s \leq t \leq T$,

$$\begin{aligned} \mathbb{E}[M_t \mid \mathcal{F}_s] &= \mathbb{E}[\mathbb{E}[X_T \mid X_t] \mid \mathcal{F}_s] \\ &= \mathbb{E}[X_T \mid \mathcal{F}_s] \\ &= P_{T-s} f(X_s) = M_s \end{aligned}$$

So M_t is a martingale. Assume μ is a probability measure. By Doob's maximal inequality,

$$\mathbb{E} \left[\sup_{t \in [0, T]} M_t^p \right] \leq C_p^p \mathbb{E}[M_T^p]$$

Because $X_0 \sim \mu$ and μ is invariant, $X_T \sim \mu$. Therefore, $M_T = f(X_T)$ and so

$$\mathbb{E}[M_T^p] = \|f\|_p^p.$$

Furthermore, by the Jensen's Inequality,

$$\mathbb{E} \left[\mathbb{E} \left[\sup_{t \in [0, T]} M_t \mid X_T \right]^p \right] \leq \mathbb{E} \left[\sup_{t \in [0, T]} M_t^p \right] \leq \|f\|_p^p.$$

By the reversibility, the law of (X_t, X_T) is the same as that of (X_{T-t}, X_0) , so

$$\begin{aligned} \mathbb{E}[M_t \mid X_T = x] &= \mathbb{E}[P_{T-t}(f)(X_t) \mid X_T = x] \\ &= \mathbb{E}[P_{T-t}(f)(X_{T-t}) \mid X_0 = x] \\ &= P_{T-t}(P_{T-t}(f))(x) = P_{2(T-t)}(f)(x), \end{aligned}$$

i.e., $\mathbb{E}[M_t \mid X_T] = P_{2(T-t)}f(X_T)$. Therefore,

$$\mathbb{E} \left[\sup_{t \in [0, T]} M_t \mid X_T \right] \geq \sup_{t \in [0, T]} \mathbb{E}[M_t \mid X_T] = \sup_{t \in [0, T]} P_{2(T-t)}f(X_T).$$

It follows that

$$\left\| \sup_{t \in [0, 2T]} P_t f \right\|_p \leq C_p \|f\|_p. \quad \square$$

When μ is symmetric, $L = L^*$ implies that

$$\partial_t p_t(x, y) = L_x p_t(x, y) = L_y p_t(x, y),$$

4. **Dirichlet Form:** Let $\mathbf{P} = (P_t)_{t \geq 0}$ be a Markov semigroup on $(E, \mathcal{F})$ with reversible measure μ and L be the Markov operator. Let Γ be the carré du champ operator on $\mathcal{A}$. Define

$$\mathcal{E}(f, g) = \int_E \Gamma(f, g) d\mu, \quad (f, g) \in \mathcal{A} \times \mathcal{A}$$

Then by the invariance and reversibility of μ ,

$$\mathcal{E}(f, g) = \int_E \Gamma(f, g) d\mu = - \int_E f L g d\mu.$$

It also implies that

$$\mathcal{E}(f, f) = - \int_E f L f d\mu \geq 0,$$

which means that $-L$ is positive.

To extend the domain of $\mathcal{E}$, first, $\mathcal{E}(f) := \mathcal{E}(f, f)$ can be defined on $\mathcal{D}(L)$. Moreover, for any $f \in \mathcal{D}(L)$,

$$\partial_t \int_E (P_t f)^2 d\mu = 2 \int_E P_t f L P_t f d\mu = -2\mathcal{E}(P_t f),$$

and

$$\begin{aligned} \partial_t \mathcal{E}(P_t f) &= -\partial_t \int_E P_t f L P_t f d\mu \\ &= - \int_E (L P_t)^2 d\mu - \int_E P_t f L^2 P_t f d\mu \\ &= -2 \int_E (L P_t f)^2 d\mu. \end{aligned}$$

So $\mathcal{E}(P_t f)$ is decreasing in t and

$$\int_E f^2 d\mu - \int_E (P_t f)^2 d\mu = 2 \int_0^t \mathcal{E}(P_s f) ds \geq 2t \mathcal{E}(P_t f)$$

Therefore, by changing t into $\frac{t}{2}$,

$$\mathcal{E}(P_t f) \leq \frac{1}{t} \left[\int_E f^2 d\mu - \int_E (P_{t/2} f)^2 d\mu \right] \leq \mathcal{E}(f)$$

Therefore, for $f \in \mathcal{D}(L)$, the middle term converges to $\mathcal{E}(f)$, as $t \rightarrow 0$. To consider a wider class, defining $\mathcal{D}(\mathcal{E})$ be the class of functions in $L^2(\mu)$ for which

$$\frac{1}{t} \left[\int_E f^2 d\mu - \int_E (P_{t/2} f)^2 d\mu \right] = \frac{1}{t} \int_E f (f - P_t f) d\mu$$

has a finite limit as $t \rightarrow 0$, which is defined as $\mathcal{E}(f)$.

Definition 1.2.6 (Dirichlet Form). For a symmetric Markov semigroup $\mathbf{P} = (P_t)_{t \geq 0}$ with reversible measure μ ,

$$\mathcal{E}(f) := \lim_{t \rightarrow 0} \frac{1}{t} \int_E f (f - P_t f) d\mu$$

for all $f \in L^2(\mu)$ for which this limit exists, and denoted by $\mathcal{D}(\mathcal{E})$. For $f, g \in \mathcal{D}(\mathcal{E})$, $\mathcal{E}(f, g)$ is defined by polarization.

Remark 1.2.7. By above, we can see, if $f, g \in \mathcal{D}(L)$,

$$\mathcal{E}(f, g) = \int_E f(-Lg) d\mu = \int_E g(-Lf) d\mu,$$

while if $f, g \in \mathcal{A}$ on which Γ is defined,

$$\mathcal{E}(f, g) = \int_E \Gamma(f, g) d\mu.$$

Hence $\mathcal{A} \subset \mathcal{D}(L) \subset \mathcal{D}(\mathcal{E}) \subset L^2(\mu)$.

Because $-L$ is self-adjoint and positive, it has the spectral decomposition

$$-L = \int_0^\infty \lambda dE(\lambda),$$

for some spectral measure $E(\lambda)$. In particular, if P_t is compact, then there exists a Hilbert basis $(e_k)_{k \in \mathbb{N}}$ of $L^2(\mu)$ such that

$$-L e_k = \lambda_k e_k, \quad k \in \mathbb{N},$$

so that

$$\mathcal{E}(e_k) = \int_E \Gamma(e_k, e_k) d\mu = - \int_E e_k L e_k d\mu = \lambda_k \int_E e_k^2 d\mu = \lambda_k$$

By the functional calculus, since $P_t = e^{tL}$,

$$P_t f = \sum_{k \in \mathbb{N}} e^{-\lambda_k t} f_k e_k,$$

for any $f = \sum_k f_k e_k \in L^2(\mu)$. Therefore,

$$p_t(x, y) = \sum_{k \in \mathbb{N}} e^{-\lambda_k t} e_k(x) e_k(y).$$

5. **Finite State Spaces:** Let E be a finite set with σ -algebra $\mathcal{F} = \mathcal{P}(E)$ and $N = |E|$. Consider a Markov generator L acting on the space of all real-valued function on E , i.e., $\mathbb{R}^N$. For the basis of $(\mathbb{1}_x)_{x \in E}$, L can be expressed as a matrix $(L(x, y))_{(x, y) \in E \times E}$.

(i) By $L\mathbb{1} = 0$, $\sum_{y \in E} L(x, y) = 0$ for any $x \in E$.

(ii) The carré du champ operator is given by

$$\Gamma(f, g)(x) = \frac{1}{2} \sum_{y \in E} L(x, y)(f(y) - f(x))(g(y) - g(x)).$$

Because $\Gamma(f) \geq 0$, $L(x, y) \geq 0$ for any $x \neq y$.

Conversely, any square matrix $L \in \mathbb{R}^{E \times E}$ with $L\mathbb{1} = 0$ and $L(x, y) \geq 0$ for any $x \neq y$ is called a Markov generator on E .

Let $P_t = e^{tL}$. Then $(P_t)_{t \geq 0}$ is the corresponding Markov semigroup, whose kernel can also be viewed as matrix $(p_t(x, y))_{(x, y) \in E \times E}$, i.e.,

$$P_t f(x) = \sum_{y \in E} f(y) p_t(x, y).$$

The heat equation reads

$$\partial_t p_t(x, y) = \sum_{y \in E} L(x, y) p_t(x, y), \quad p_0(x, y) = \delta_x(y).$$

A finite measure μ is invariant with respect to L if and only if for all $y \in E$,

$$\sum_{x \in E} \mu(x) L(x, y) = 0.$$

So it is an eigenvector of $L^\top$ with respect to eigenvalue 0 and it always exists because L is not invertible. So such invariant probability measure μ is unique if and only if $\ker L^\top = 1$. Moreover, if L is irreducible, that is, for any $(x, y) \in E \times E$, there exists a path $(x_0 = x, x_1, \dots, x_\ell = y)$, such that $L(x_i, x_{i+1}) > 0$, then such μ is also unique, because $\dim \ker L^\top =$ the number of closed communicating classes.

Furthermore, if μ is reversible, then

$$\mu(x) L(x, y) = \mu(y) L(y, x).$$

Therefore, by defining $\mu(x_0) = 1$ for some $x_0 \in E$, let

$$\mu(x) = \frac{L(x_0, x)}{L(x, x_0)}, \quad \forall x \in E.$$

Then the invariant measure can be obtained by normalizing μ . Furthermore, when L is irreducible, $\mu(x) > 0$ for $x \in E$, and

$$R(x, y) = \sqrt{\mu(x)} L(x, y) \frac{1}{\sqrt{\mu(y)}}$$

is a symmetric matrix (so is L).

1.3 SDE and Diffusion Process

1. **Diffusion Process:** Consider SDE

$$dX_t^x = \sigma(X_t^x) dB_t + b(X_t^x) dt, \quad X_0^x = x,$$

on $\mathbb{R}^n$, where σ and b are smooth with bounded derivatives. Then a process $\mathbf{X} = \{X_t^x; t \geq 0, x \in \mathbb{R}^n\}$ which is a solution of above is called a diffusion process. Then by the Itô's formula, for $f \in C^2(\mathbb{R}^n)$,

$$df(X_t^x) = dM_t^f + Lf(X_t^x) dt, \quad X_0^x = x$$

where

$$dM_t^f = \sum_{i,j=1}^n \sigma_i^j(X_t^x) \partial_j f(X_t^x) dB_t^i,$$

and

$$Lf = \frac{1}{2} \sum_{i,j=1}^n g^{ij} \partial_{ij}^2 f + \sum_{i=1}^n b^i \partial_i f$$

for $g^{ij} = \sum_k \sigma_k^i \sigma_k^j$, i.e., $\mathbf{g} = (g^{ij}) = \sigma \sigma^\top$. Furthermore, such L is the corresponding Markov generator of $\mathbf{X}$, which can be proved by the Dynkin's formula. Note that $(M_t^f)_{t \geq 0}$ is a local martingale. When f and Lf is bounded, it is a true martingale and then

$$\mathbb{E}(f(X_t^x)) = f(x) + \int_0^t \mathbb{E}(Lf(X_s^x)) ds.$$

Furthermore,

$$\Gamma(f, g) = \sum_{i,j=1}^n g^{ij} \partial_i f \partial_j g.$$

2. **Diffusion Operator:** An operator L , with carré du champ operator Γ , is said to be a diffusion operator or generator if

$$L\psi(f) = \psi'(f)Lf + \psi''(f)\Gamma(f)$$

for $\psi \in C^2(\mathbb{R})$ and suitably smooth function f .

Equivalently,